MACHINE LEARNING WITH PYTHON PROGRAMMING

COMPREHENSIVE HANDS-ON PRACTICAL TEST

Title: Comparative Analysis of Machine Learning Classification Algorithms on a Real-World Dataset

Duration: 60 Minutes **Maximum Marks:** 100 **Mode:** Individual Practical Examination

1. Objective

Develop an end-to-end classification pipeline, implement multiple machine learning models, compare their performance using appropriate metrics, and interpret the results. This assessment tests data preparation, model selection, dimensionality reduction, hyperparameter tuning, and evaluation.

2. Dataset

Breast Cancer Wisconsin (Diagnostic) — binary classification of malignant and benign tumors.

Dataset hyperlink: https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

Property	Description
Samples	569
Features	30 numerical features
Classes	2 (Malignant and Benign)
Missing values	None in the Scikit-learn version

Load the dataset: from sklearn.datasets import load_breast_cancer; then run data = load_breast_cancer(as_frame=True), X = data.data, and y = data.target.

3. Practical Tasks

Task 1: Data Exploration and Preprocessing (15 Marks | 10 Minutes)

• Display dataset shape, feature names, first five records, missing-value counts, and class distribution.

• Create stratified splits: 70% training, 15% validation, and 15% testing; use random_state=42.

• Apply StandardScaler where appropriate. Fit only on training data, then transform validation/test data.

Required outputs: dataset summary, split shapes, and class-distribution bar chart.

Task 2: Implement Six Model Configurations (35 Marks | 20 Minutes)

ID	Algorithm	Configuration
M1	Logistic Regression	max_iter=2000
M2	K-Nearest Neighbors	n_neighbors=5
M3	Gaussian Naive Bayes	Default parameters
M4	Decision Tree (Gini)	criterion="gini"
M5	Decision Tree (Entropy)	criterion="entropy"
M6	PCA + KNN	PCA(n_components=0.95), then KNN (k=5)

• Train all models with the same training split; generate validation and test predictions.

• Standardize before PCA; fit PCA on the training set only. Report the number of retained components.

• Measure fitting time and test-set prediction time separately using time.perf_counter().

Note: Gaussian Naive Bayes is appropriate for continuous features. Linear Regression is excluded because the current task is classification.

Task 3: Evaluation and Visualizations (30 Marks | 15 Minutes)

Evaluate all six configurations on the **same held-out test set**. Report Accuracy, macro Precision, macro Recall, macro F1-score, training time (seconds), and prediction time (milliseconds).

Mandatory comparison table:

Model	Accuracy	Precision	Recall	F1	Train (s)	Predict (ms)
Logistic Regression						
KNN						
Gaussian Naive Bayes						
Decision Tree (Gini)						
Decision Tree (Entropy)						
PCA + KNN						

• Create a bar chart comparing the test F1-scores of all six models.

• Show confusion matrices for the highest- and lowest-F1 models. Label figures clearly.

Task 4: Hyperparameter Analysis and Interpretation (20 Marks | 15 Minutes)

Part A — KNN hyperparameter study (10 marks): Test k = [1, 3, 5, 7, 9]. Plot validation accuracy vs. k, select the best k using the validation set, and report its final test accuracy.

Part B — Short answers (10 marks):

• Which model has the highest test F1-score? Is it also the fastest?

• How does PCA change KNN accuracy and prediction time? How many principal components are retained?

• Do Gini and Entropy trees produce identical results? Explain.

• Why does scaling matter for KNN and Logistic Regression, but generally not for Decision Trees?

• Considering predictive quality and speed, which model would you select? Justify using results.

4. General Instructions

• Work individually and finish within 60 minutes.

• You may use NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn.

• Use random_state=42 where applicable and identical data splits across models.

• Select hyperparameters using validation data, never the held-out test data.

• Results must be computed by running code; do not fabricate values.

• Label tables and plots; explain observations based on evidence.

5. Submission Guidelines

Submit one executed Jupyter Notebook named **RollNo_ML_Practical_Test.ipynb**, containing the preprocessing, all six models, performance table, visualizations, KNN experiment, and five short-answer responses. Keep outputs visible.

6. Evaluation Rubric

Assessment component	**Marks**
Dataset loading, exploration, splitting, and preprocessing	15
Six model implementations	30
PCA integration	5
Evaluation metrics and comparison table	20
Performance visualizations	10
KNN hyperparameter experiment	10
Critical analysis and interpretation	10
TOTAL	100